Q05: When Does Extra Reasoning Help?

Test-time compute and its trade-offs

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The question

Student question (#05)
When does extra reasoning actually improve an answer, and what is the trade-off?

What “extra reasoning” is

- Chain-of-thought: emit intermediate steps.
- Self-consistency / best-of-n: sample many, aggregate or verify.
- Reasoning models: trained to produce long deliberate traces.

All trade tokens (latency, \$) for a *chance* at accuracy.

When it helps

- **Multi-step** tasks (math, code, logic, planning).
- **Verifiable** answers $\Rightarrow$ sampling + selection pays off.
- Test-time compute can *substitute* for parameters.

When it doesn't—or hurts

- Simple/lookup tasks: tokens wasted, can talk itself out of a right answer.
- **Overthinking**: accuracy rises then falls (inverted-U).
- **Faithfulness gap**: the trace may not be the real computation.

The trade-off, quantitatively

Coverage of best-of-n with single-sample accuracy p_1:

$$\Pr[\text{at least one correct}] = 1 - (1 - p_1)^n$$

- Cost $\sim$ linear in n (and trace length).
- Returns **diminish** $\Rightarrow$ an optimal budget per query.
- Requires a verifier to *capture* the coverage.

A decision rule

Spend extra reasoning when:

1. the task is genuinely multi-step,
2. errors are costly,
3. you can verify / cross-check candidates.

Otherwise keep it short. Calibrate per task—never one length globally.